

Biodegradation of the Polyurethane Foam Covering of Breast Implants

Thomas M. Sinclair, M.D., Carolyn L. Kerrigan, M.D., F.R.C.S.(C), and Rudolf Buntic, M.D.

Montreal, Quebec, Canada

Although it is generally accepted that polyurethane-covered breast implants have decreased the incidence of clinical capsular contracture, there remain many unanswered questions regarding the physical and chemical degradation of the polyurethane foam covering itself. We have systematically studied the fibrous capsule and polyurethane foam recovered from human breast "explants" in an effort to characterize more precisely the biodegradation of polyurethane foam in the human body.

Seventy-five freshly retrieved polyurethane-covered implants and surrounding capsule from 47 patients have been analyzed. Capsular tissue from several sampling sites around the surface of the implants was digested in a collagenase solution until foam was recovered or all tissue was digested. Additional samples were fixed in 10% formalin. Scanning electron microscopy was used to look for structural changes in the recovered intact foam and to determine the foam strut widths. Fourier transform IR spectroscopy and x-ray photoelectron spectroscopy were used to analyze the chemical composition of the polyurethane. The formalin-preserved capsule samples were examined histologically for further evidence of foam degradation.

Of the 75 prostheses analyzed, 36 (48 percent) were removed because of capsular contracture and 10 (13 percent) because of infection or exposure of the prosthesis. The remaining 29 (39 percent) implants were removed for various other reasons. Visibly intact foam was recovered from 36 (48 percent) prostheses after enzymatic digestion of capsule tissue. There was a progressive decline in the ability to recover intact foam as the total implantation time increased. Scanning electron microscopy revealed fractures and fissures in the foam structure and thinning of the polyurethane struts. The mean strut width of control, unimplanted foam was $49 \pm 1.5 \mu\text{m}$ ($\pm\text{SEM}$). Retrieved foam from implants which developed capsular contracture and the infected implants had strut widths of 30 ± 3.1 and $32 \pm 3.1 \mu\text{m}$, respectively. In implants removed for other reasons, the

polyurethane foam strut width was $41.2 \pm 2.3 \mu\text{m}$. Despite an inability to recover visibly intact foam from 39 specimens, standard light microscopy of 37 of these same specimens showed residual polyurethane still present in the capsule. Various degrees of scalloping and fracturing of the foam were seen in the histologic sections.

There is convincing evidence by scanning electron microscopy and histology that polyurethane is degrading. It was not possible to quantitate accurately the rate of degradation, but factors such as capsular contracture, infection, and time appear to have a role in the biodegradation of polyurethane in the human body. These relationships require further study. (*Plast. Reconstr. Surg.* 92: 1003, 1993.)

Frustrations with high clinical rates of capsular contracture led Ashley¹ to use a polyurethane foam-covered breast prosthesis in the early 1970s. As clinical experience with this type of implant increased, it became apparent that this modified implant design significantly reduced the incidence of clinical capsular contracture, with many surgeons reporting incidence rates as low as 1 to 5 percent.²⁻⁵ Many investigators have studied the tissue reaction to the polyurethane in an attempt to elucidate the mechanism behind the successful decrease in capsular contracture. One of the more popular theories is that the open cell structure of the foam prevents collagen from aligning uniformly around the implant, thus altering the forces involved in spherical contraction.⁶⁻⁸ The mechanism is not completely understood, but what is certain is that a prolonged and intense foreign-body reaction oc-

From the Microsurgical Research Laboratories at the Royal Victoria Hospital and McGill University. Received for publication December 10, 1991; revised November 19, 1992.

This work was presented, in part, at the 36th Annual Meeting of the Plastic Surgery Research Council, in Charlottesville, Virginia, on May 6, 1991, and at the 60th Annual Meeting of the American Society of Plastic and Reconstructive Surgeons, in Seattle, Washington, on September 23, 1991.

curs and macrophages fuse into multinucleated giant cells in response to the presence of the foam.⁷⁻¹¹

This prolonged inflammatory response is also believed to be responsible, in part, for the degradation of the polyurethane foam.⁸ Reports in the literature have documented evidence of microscopic degradation and fragmentation of the foam,^{7,8,12,13} as well as clinical anecdotes claiming disappearance of the foam covering.^{14,15} These findings have raised many concerns about the implant's safety and biocompatibility as a long-term implantable device. Most of the evidence, however, has been an interpretation of standard light microscopic analysis of the capsule tissue; thus there exists little documentation or understanding of the actual structural or chemical changes that occur in the polyurethane foam.

The purpose of this study, using freshly explanted human polyurethane foam-covered breast prostheses, was to try to characterize more precisely the biodegradation of the polyurethane foam in the human body (1) by documenting the physical and chemical changes that occur in the foam over time and (2) by analyzing and comparing the foam from explants of patients with uncomplicated and complicated post-operative clinical courses. It should be noted that most of the implants received were removed because of problems; therefore, interpretation of the results must be made within this context.

MATERIALS AND METHODS

A mailing was sent to members of the Association des Chirugiens Plasticiens du Québec and the Canadian Society of Plastic Surgeons which informed them of our research interest and requested that they send freshly explanted polyurethane foam-covered breast prostheses and the surrounding tissue capsule to our institution for analysis. The surgeons were instructed to ship the specimens in saline and packed on ice and to complete a detailed questionnaire. This questionnaire included documentation of the surgical technique, the complications, the reasons for implantation and removal, the dates of surgeries, the manufacturer's lot number, and any additional comments.

When implants with intact surrounding capsules were received, capsular tissue from approximately eight random sampling sites was removed by first incising the capsule and then carefully peeling the tissue off the underlying

silicone elastomer shell. If the capsule was already separated from the implant, care was taken to sample an area that clearly showed adipose tissue present on one side and a smooth, glistening, fibrous surface on the other. This ensured that the sampling sites (1 × 2 cm biopsy) represented a full thickness of capsular tissue. A 3-mm-wide strip of capsule tissue from the middle of each biopsy specimen was cut out and then fixed in buffered 10% formalin (Fig. 1).

The biopsy specimens were then placed in 20 ml of a solution of crude collagenase isolated from *Clostridium histolyticum* (Sigma Chemical Company, St. Louis, Mo.). The collagenase solution was prepared as a 0.005% w/v concentration by dissolving the collagenase into distilled water buffered with sodium monobasic phosphate (Fisher Scientific Company, Fairlawn, N.J.) to a pH of 7.4.

Eight implants with completely intact capsules and total implantation periods of 7, 7, 20, 28, 38, 39, 39, and 60 months underwent "total capsule dissolution." After a 1 × 1 cm biopsy of the capsule tissue was dissected off the implant and fixed in 10% formalin, the entire implant and surrounding capsule were immersed in 300 ml of the collagenase solution.

The capsule biopsy specimens and the eight complete capsules placed in the collagenase solution were incubated at 37°C until the polyurethane foam covering was retrieved or all the tissue was digested, whichever came first. If any *visibly intact* foam was recovered after capsule digestion, it was thoroughly but gently irrigated with sterile water to remove any small pieces of tissue remaining in the porous structure. The samples were then left to dry at room temperature. If visibly intact polyurethane foam was not recovered from the digested capsule, then the residual collagenase fluid was centrifuged, the supernatant discarded, and 0.1 ml of the remaining sediment was pipetted onto a glass slide. This

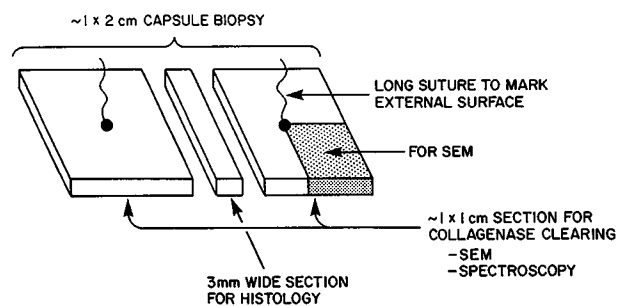


FIG. 1. Processing technique for each capsule biopsy specimen.

preparation was left to dry at room temperature and then stained with Giemsa.

Seven 1 × 1 cm control samples of polyurethane foam were dissected off new prostheses. Samples of the virgin control foam underwent processing in collagenase and sterile water in the same manner as the explanted foam.

Analysis

Scanning Electron Microscopy. All control and recovered "intact" polyurethane foam was prepared as 0.5 × 0.5 cm pieces for scanning electron microscopy by mounting the sample on an aluminum post with liquid graphite. Care was taken to ensure that the external surface of the polyurethane foam was face-up when mounted. Sputter-plating of the polyurethane samples with gold was then performed using a Hummer VI Sputter System (Anatech Ltd.) for a duration of 4 minutes at a plasma discharge current of 15 mA. Scanning electron microscopy, using a JEOL JSM-840A electron microscope with an accelerating voltage set at 15 kV, was then performed. Images of the samples were photographed with Polaroid film (type 55) at magnifications of 60×, 75×, 200×, and 1000×.

These photographs were used to record qualitative structural changes of the polyurethane struts, as well as any distortion of the open cell network. The 75× photograph for each sample was analyzed by measuring the width of each of 20 struts at its thinnest portion. The average strut width was determined for each sample.

Spectroscopy. Control and recovered polyurethane foam samples were sent to the Bureau of Radiation and Medical Devices, Health and Welfare, Canada, for both bulk and surface chemical analysis. These results will be the subject of a separate report.

Light Microscopy/Histology. Formalin-fixed capsule specimens were embedded in paraffin, sectioned, stained with Giemsa, and examined histologically. Particular note was made of the presence of foam, the presence of a foreign-body inflammatory response, and the presence of structural changes in the foam. The Giemsa-stained collagenase sediment specimens also were examined microscopically to look for foam particles.

RESULTS

Ninety-one polyurethane foam-covered breast prostheses were sent to our institution. Seventy-five specimens from 47 patients were

found suitable to analyze completely. Sixteen specimens were excluded from the study because either the clinical information was incomplete ($n = 7$), the capsule specimen was shipped in formalin ($n = 5$), or it was too small to analyze adequately ($n = 4$). Fifty-two of the implants removed were Meme models, 14 were Replicon, 5 were Optimam, and the four oldest implants, all removed about 17 years after initial insertion, were the Natural-Y model of polyurethane-covered implant.

Table I lists the reasons why the prostheses were initially inserted and subsequently removed. The "Other" group represents prostheses removed for size change ($n = 10$), patient concern over health risks ($n = 8$), hematoma ($n = 3$), malposition ($n = 3$), pain ($n = 2$), benign breast masses ($n = 2$), or breast cancer ($n = 1$). Forty-eight percent of the implants were removed because of capsular contracture. Eight of the 10 implants removed for infection/exposure had documented bacterial contamination with *Staphylococcus aureus*. Bacterial culture information from the other two implants could not be obtained.

The implantation time for each of the three groups is shown in Figure 2. The implants that developed capsular contracture had the longest implantation time. While the implantation time for the prostheses that developed contracture ranged from 5 months to 17 years, the onset of clinically evident capsular contracture ranged from 3 months to 1 year prior to removal of these implants. Visibly intact foam was recovered from 48 percent (36 of 75) of the implants after collagenase digestion of the capsule tissue. In the remaining 52 percent (39 of 75), enough alteration in structural and mechanical integrity had occurred to make intact foam unretrievable. It must be noted, however, that polyurethane was still present microscopically in 37 of these 39 capsules (see *Histology* results).

In the prostheses subjected to "total capsule dissolution" (Fig. 3), a portion of the poly-

TABLE I
The Reasons for Implant Insertion and Removal

Reason for Insertion	Reason for Removal		
	Capsular Contracture	Infection/Exposure	Other
Augmentation ($n = 57$)	30	3	24
Reconstruction ($n = 7$)	0	2	5
Subcutaneous mastectomy ($n = 11$)	6	5	0

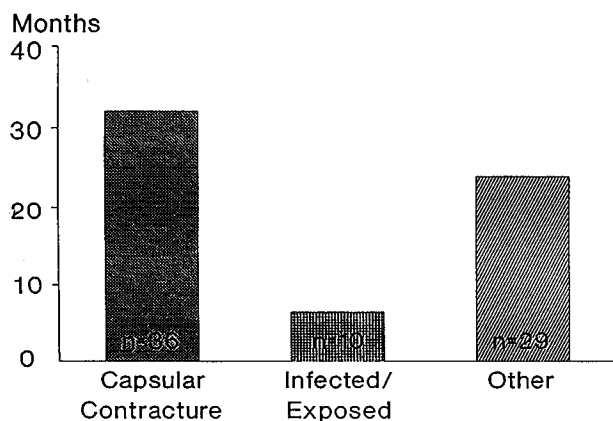


FIG. 2. Median implantation time for each group of implants. Range: Capsular contracture, 5 months to 17 years; Infected/Exposed, 1.5 months to 3 years; Other, 2 weeks to 17 years.

urethane foam covering was recovered intact from the four implants with the shortest implantation time (7, 7, 20, and 28 months). Visibly intact foam was not retrieved from the other four prostheses. There also was a progressive decline in the ability to recover *visibly intact* foam from the capsule biopsy specimens as the implantation time increased (Fig. 4). The implantation time ranged from 2 to 17 years for the 39 implants in which no intact foam was recovered. Twenty-seven of these 39 implants were removed because of capsular contracture. The other 12 were removed because of patient concern over health risks ($n = 4$), a change in implant size ($n = 2$), undiagnosed breast pain ($n = 2$), benign masses found in the breast ($n = 2$), malposition ($n = 1$), and exposure ($n = 1$).

Multiple logistic regression was performed to analyze the effects of time and reason for implantation on the probability of retrieving intact

polyurethane foam from an implant (Table II). There was a significant negative association between duration of implantation and the likelihood of retrieving intact polyurethane foam. The data showed that the probability of retrieving foam decreased by 16 percent for every month that the implant remained in place. Moreover, after adjusting for implantation time, the probability of retrieving intact polyurethane foam decreased by 93 percent for implants that developed capsular contracture as compared with implants removed for other reasons. Statistical analyses could not be performed on the infected/exposed group of implants because of too small a sample size.

Scanning Electron Microscopy

Scanning electron micrographs of control foam (Fig. 5) demonstrated a very uniform structure of polyurethane struts joining together at nodes to produce a regular pore configuration. The polyurethane foam struts were smooth, with no evidence of fractures or any other signs of surface erosion present. Intact foam was recovered from only 9 of the 36 implants that developed capsular contracture. The micrographs of these specimens showed the basic structural architecture to be unaltered, although it was somewhat distorted. In some areas, the structure appeared collapsed and compressed, and there also was a noticeable thinning of many of the foam struts in the specimens (Fig. 6). Foam was recovered from 9 of 10 of the infected/exposed implants, and these specimens all showed similar changes by scanning electron microscopy. The basic architecture again re-

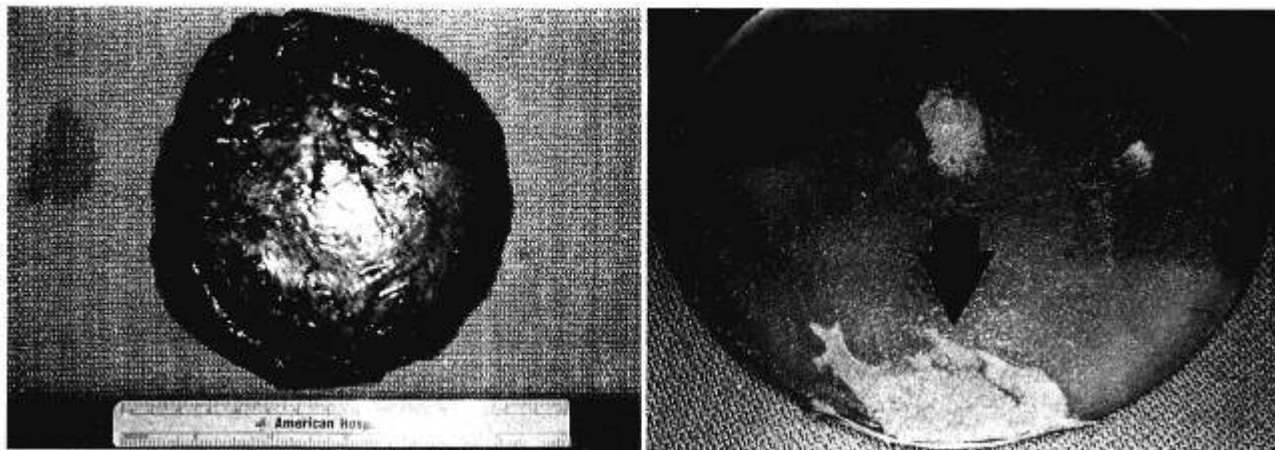


FIG. 3. Polyurethane-covered implant before (left) and after (right) total capsule dissolution by collagenase. Implant was removed 28 months after insertion for size change. Note the small amount of intact foam recovered (arrows).

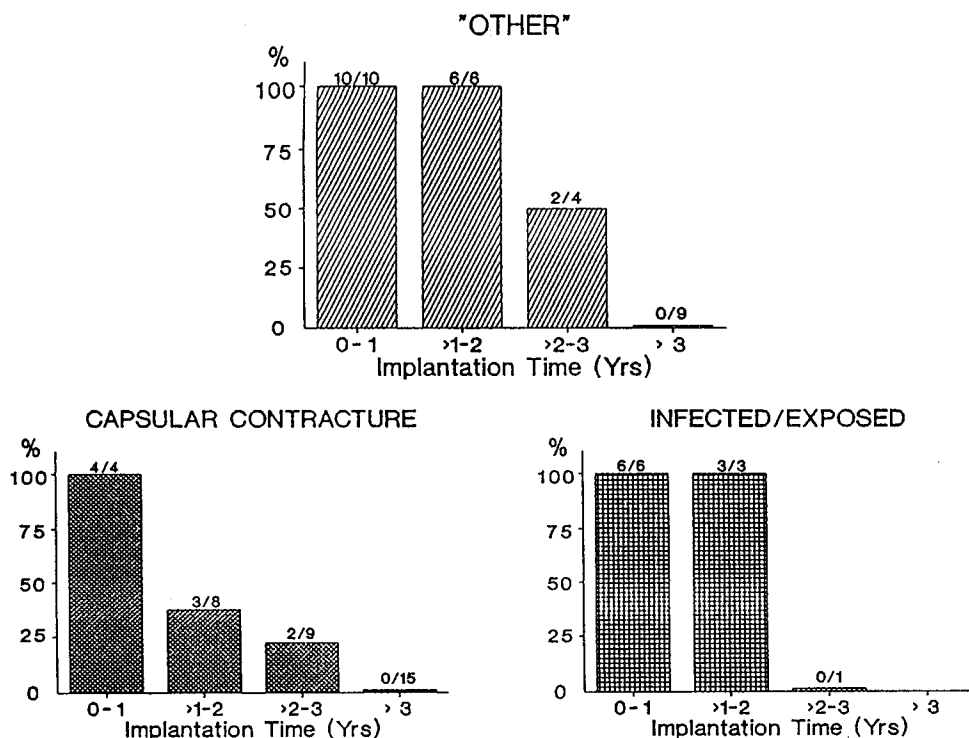


FIG. 4. The percentage of implants, in each group, with visibly intact polyurethane recovered versus the implantation time.

TABLE II
Multiple Logistic Regression for Odds of Retrieving
Intact Polyurethane Foam

Covariate	Parameter Estimate	Odds Ratio	95% Confidence Interval
Time	-0.17	0.84*	0.77-0.92
Reason for implantation:			
Contracture	-2.60	0.07*	0.01-0.52
Infection	12.7	NC†	NC

* $p < 0.0001$.

† NC: parameter could not be calculated because of zero cell.

mained intact, but this group of implants showed measurable thinning of the struts in all the foam samples, and fractures and fissures in the struts were evident (Fig. 7). The "Other" group of implants showed the greatest variation in findings. Strut width and pore architecture were generally preserved, but fracture lines and fissures in the struts were seen (Fig. 8). Some of these findings were evident even in specimens removed within 5 days after implantation. Four prostheses demonstrated extensive changes in the polyurethane struts despite a benign clinical course.

The average strut width was calculated for control foam and for the three groups of explants (Fig. 9). Recovered foam from the implants that developed capsular contracture and

the infected implants had the thinnest polyurethane struts, with mean widths of $30.2 \pm 3.1 \mu\text{m}$ ($\pm\text{SEM}$) and $32.1 \pm 3.1 \mu\text{m}$, respectively. The "Other" group had an average strut width of $41.2 \pm 2.3 \mu\text{m}$, while the control foam had the thickest polyurethane struts, measuring $48.9 \pm 1.5 \mu\text{m}$. Microstructural changes in the enzymatically recovered polyurethane were evident from all the explants, but the open-pore structure always remained intact.

Light Microscopy/Histology

Light microscopy of the collagenase sediment from implants in which visibly intact foam was not recovered enzymatically revealed large numbers of foam particles still present (Fig. 10). All the particles were too small to see with the naked eye. Although the shapes of the foam particles varied, most of the particles were either rectangular or triangular. Fracture lines and blunted or jagged edges were seen in some particles.

The histologic sections of the capsules from implants in which intact foam was recovered enzymatically showed the typical triangular appearance of the struts and the hourglass appearance of the nodes when they were cut in cross section (Fig. 11). Abundant foam was present in these samples, and an intense foreign-body in-

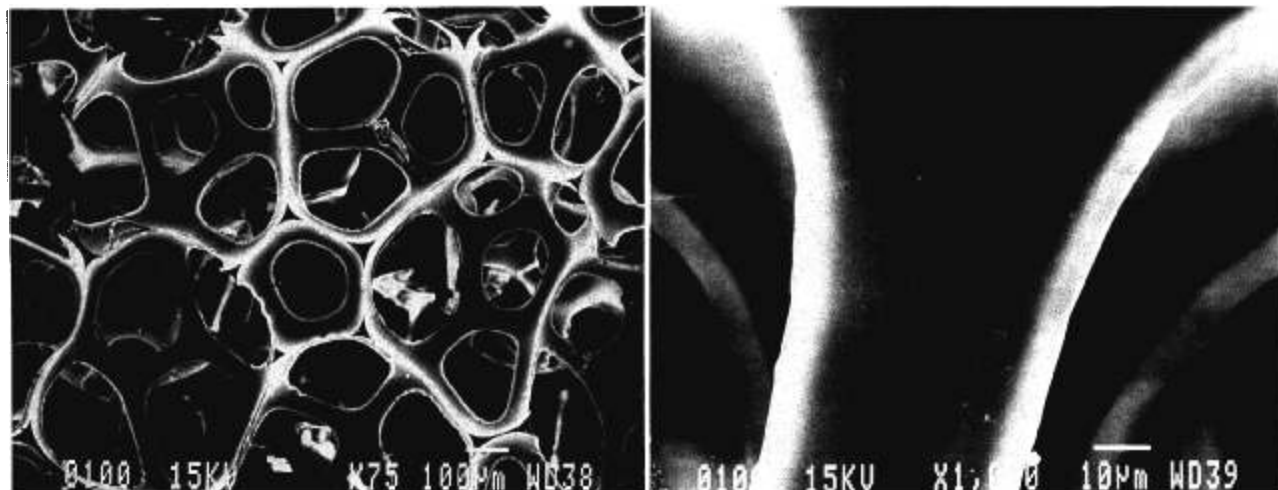


FIG. 5. Scanning electron microscopy of control foam. Regular open-pore structure (left; $\times 75$) with uniform, smooth-surfaced struts (right; $\times 1000$).

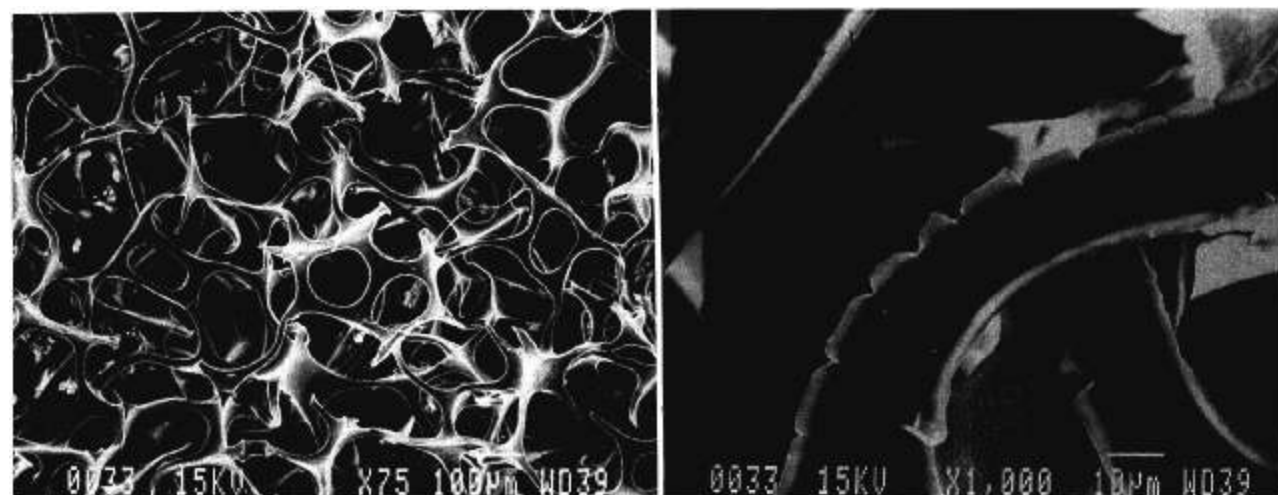


FIG. 6. Polyurethane recovered from implant removed after 13 months because of capsular contracture. Distortion of structure is evident, and the struts are thinned (left; $\times 75$) with surface defects present (right; $\times 1000$).

flammatory response also was present. Many multinucleated giant cells were seen surrounding the foam struts. The foam showed some fracturing, but the edges of most of the struts were sharp and well demarcated, indicating that little, if any, structural change had occurred. These findings are in sharp contrast to what was found in the capsules of implants in which intact foam could *not* be recovered by enzymatic digestion. The foam from these implants varied in its appearance from capsule to capsule and from site to site within the same capsule. Some of the capsules had abundant foam present, while other capsules were very sparse in foam (Fig. 12). The implants with the longest implantation time tended to have the least amount of foam present; however, this was not a consistent finding. The foam struts appeared smaller than those seen in

the capsules of implants in which foam was recovered. The same foreign-body response, with multinucleated giant cells surrounding the foam struts, was evident. Blunting of the foam edges, however, was a very common finding, with fractures and scalloped edges present in many of the specimens as well (Fig. 13). There was a marked decrease in the foreign-body inflammatory response as the total implantation time increased. Many of the older capsules consisted primarily of fibrous tissue with no multinucleated giant cells or inflammation present.

All the capsule specimens examined, except those which came from two prostheses removed 17 years after implantation, had microscopic evidence of foam still present in the capsule, even though visibly intact foam was not recovered by collagenase digestion of the capsule tissue.

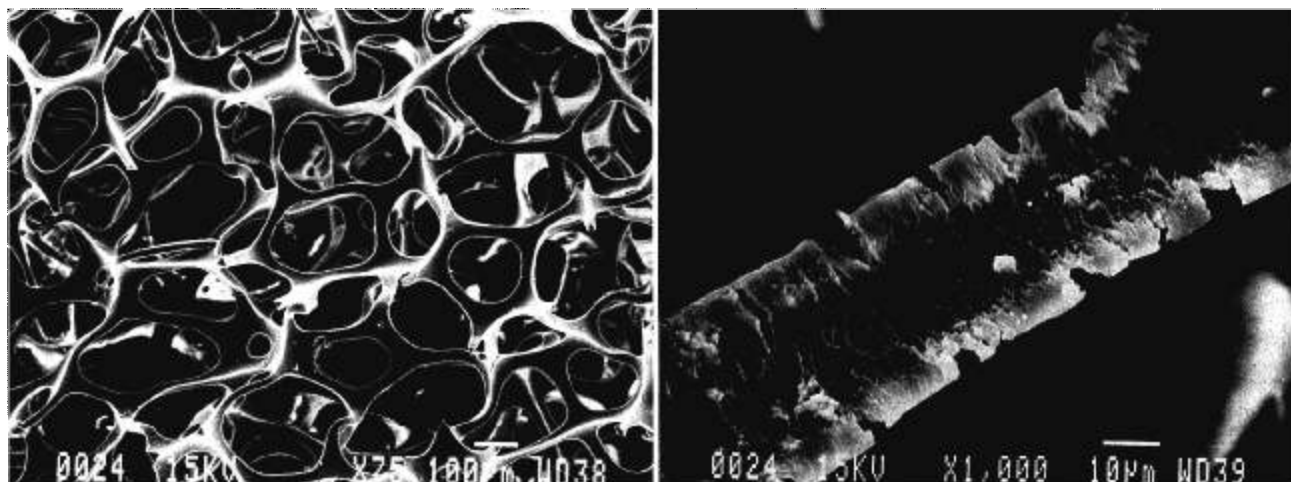


FIG. 7. Polyurethane recovered from an infected implant removed after 6 months. Again, structure is intact with thinned struts (left; $\times 75$), but extensive surface fissuring and fracturing is seen (right; $\times 1000$).

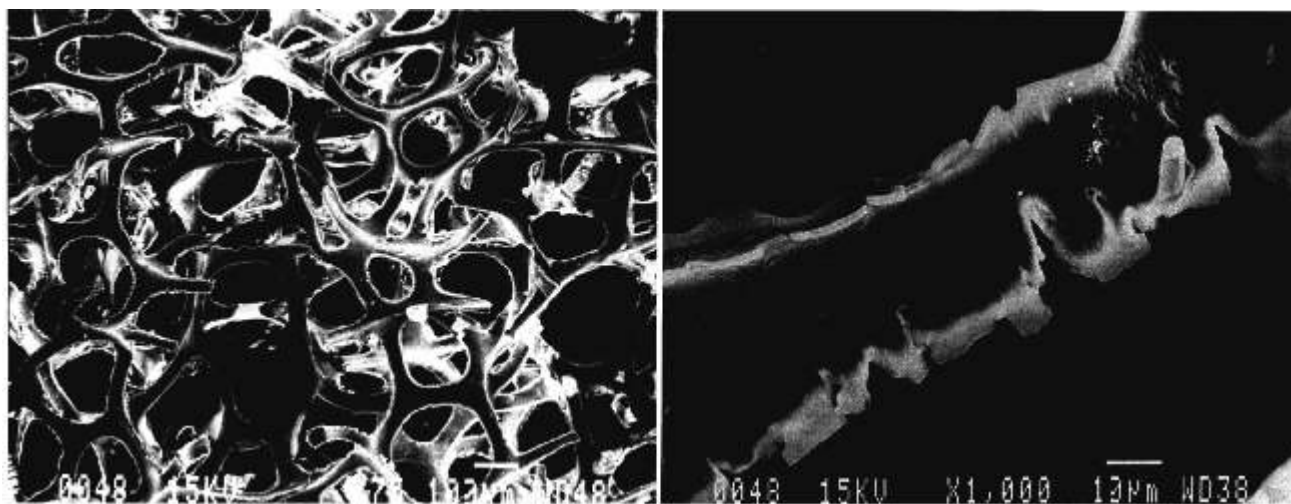


FIG. 8. Polyurethane recovered from implant removed after 28 months (see Fig. 3, right) for size change. Note that the basic structural network is unaltered (left; $\times 75$), but surface defects are again seen (right; $\times 1000$).

DISCUSSION

Biodegradation has been assigned various definitions in the literature,¹⁶ and we have adopted the definition proposed by Marchant et al.,¹⁷ who presented biodegradation "as occurring at many different structural levels, i.e., molecular, macromolecular, microscopic, and macroscopic, depending on the mechanism." It also must be remembered that polyurethane is not a single material but a family of compounds with widely different physical, chemical, and mechanical properties.¹⁸ The precise formulation of the polyurethane foam used in breast implants and thus analyzed in this study is proprietary information, but it is known that the polymer is a toluene diisocyanate-based polyesterurethane.¹⁹

The tissue reaction to the polyurethane foam

covering has been well described previously^{8,13} and will not be discussed in depth here. As other authors have demonstrated, a prolonged foreign-body inflammatory response develops with multinucleated giant cells surrounding the polyurethane struts. In addition, we have observed that as the implantation time increases, a more fibrous reaction gradually replaces the inflammatory tissue reaction such that the capsule almost becomes histologically indistinguishable from capsules seen around smooth-surfaced silicone implants. The older capsules also appeared to have smaller and fewer foam particles which were more widely dispersed, implying a gradual biodegradation of the foam. This was not a universal finding, however, and quite clearly there is variability not only from patient to patient but from site to site within the capsule as

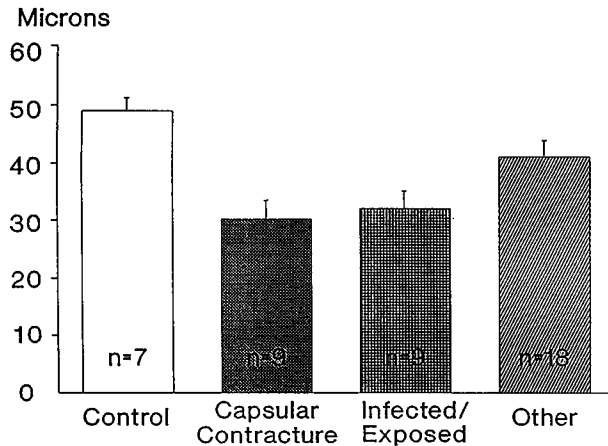


FIG. 9. The average polyurethane strut width in microns (mean $\pm$ SEM) for control foam and in foam recovered intact after collagenase digestion of capsules.

well. Thus we could not devise an effective technique to quantitate the residual foam without fear of a significant sampling error.

The mechanism of polyurethane biodegradation is likely multifactorial, involving mechanical stress forces^{13,20} produced during capsule formation, as well as enzymatic, oxidative, and hydrolytic processes from surrounding inflammatory cells during the foreign-body reaction.^{16,20,21} Structural changes in the polyurethane, however, have not been well documented and are based primarily on light microscopic analysis of capsule tissue. We found that Giemsa stain, which stains polyurethane a deep blue, provided a greater detail of the foam particles than conventional hematoxylin and eosin or trichrome stains. We also noted what other authors have described as "migration of foam particles" and "fragmentation of foam particles,"^{8,13} but are these true findings or merely artifacts from histologic processing? Scanning electron microscopy of polyurethane foam recovered by enzymatic capsule digestion enabled us to correlate actual structural changes in the polyurethane with the microscopic findings seen by light microscopy. Scanning electron microscopy was a more sensitive method than light microscopy for demonstrating subtle surface erosions, fissures, and fractures; however, it could only be used to analyze polyurethane foam whose macroscopic structure remained intact after the capsule was digested enzymatically.

Although Szycher and Siciliano²² were able to recover some intact polyurethane foam from a 9-year implant, we could not recover intact foam from any implant older than 3 years. Even total capsule dissolution of a 28-month implant did

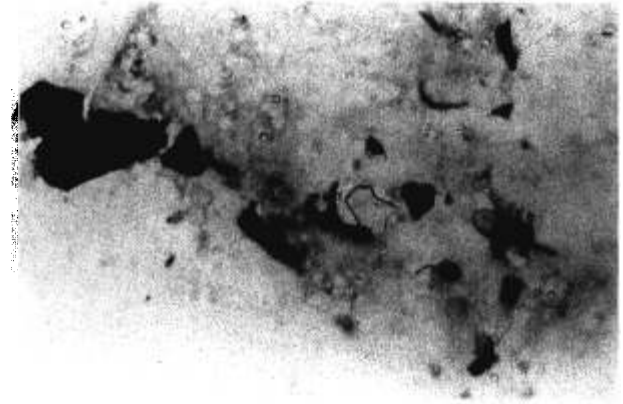


FIG. 10. Sediment from centrifuged collagenase fluid when digestion of capsule tissue did not yield intact foam. Many foam particles are present (Giemsa stain; $\times 250$).

not yield more than a 3×3 cm piece of intact foam, which was approximately 5 percent of the total foam covering. Because of this discrepancy between Szycher and Siciliano's findings and our own, we resampled capsule tissue from 10 implants, in 9 of which we had been unable to recover intact foam after collagenase digestion but by histology showed foam particles to be present in the capsule. The implantation time ranged from 2 to 6 years for these prostheses. We subsequently subjected the pieces of capsule tissue to the same enzyme solution (Polymedica Industries, Woburn, Mass.) used by Szycher and Siciliano. The results were exactly the same as our own, in that we again could not retrieve intact foam from these same 9 implants but did recover foam, as expected, from the other implant.

Over half the implants in our own study did not yield any intact foam after enzymatic digestion of the capsule, yet light microscopic examination of many of these same capsule specimens showed a large amount of foam still present. This suggests that the polyurethane struts are gradually thinned until the foam network collapses secondary to loss of its macroscopic structural integrity. The fate of the remaining microscopic particles of foam is open to speculation. It was therefore impossible to determine accurately the rate of polyurethane degradation for any given implant because of the inability to recover and quantitate the amount of foam present as microscopic particles in the local tissue.

Statistically, we were able to demonstrate that duration of implantation and the presence of capsular contracture were significant factors contributing to the degradation of the foam

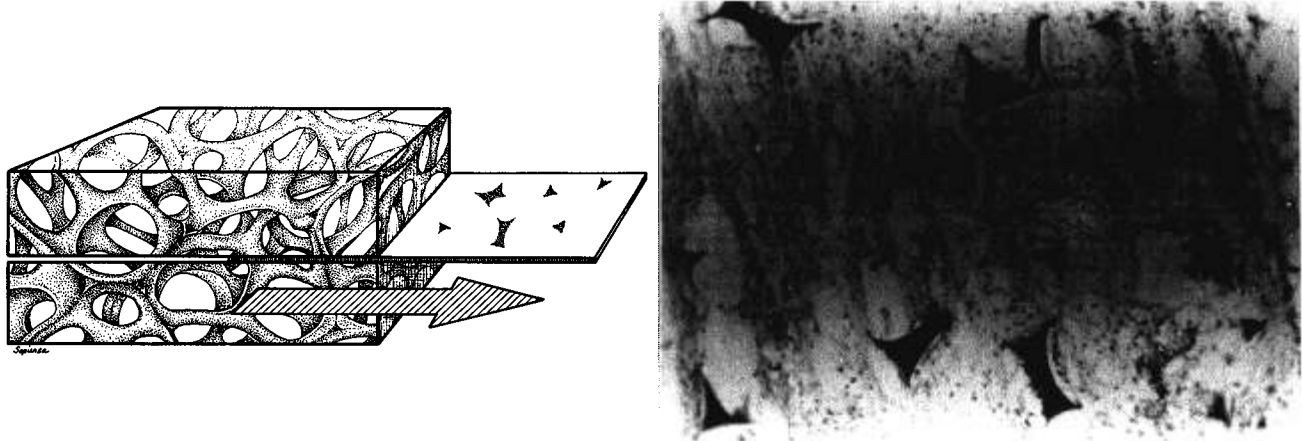


FIG. 11. (Left) Schematic representation of polyurethane foam cut in cross section demonstrating three-sided struts and four-sided nodes. (Right) Typical histology of a capsule in which intact polyurethane foam would be recovered by collagenase digestion (Giemsa stain; $\times 100$). Multinucleated giant cells surround the foam, but there is no gross evidence of degradation.

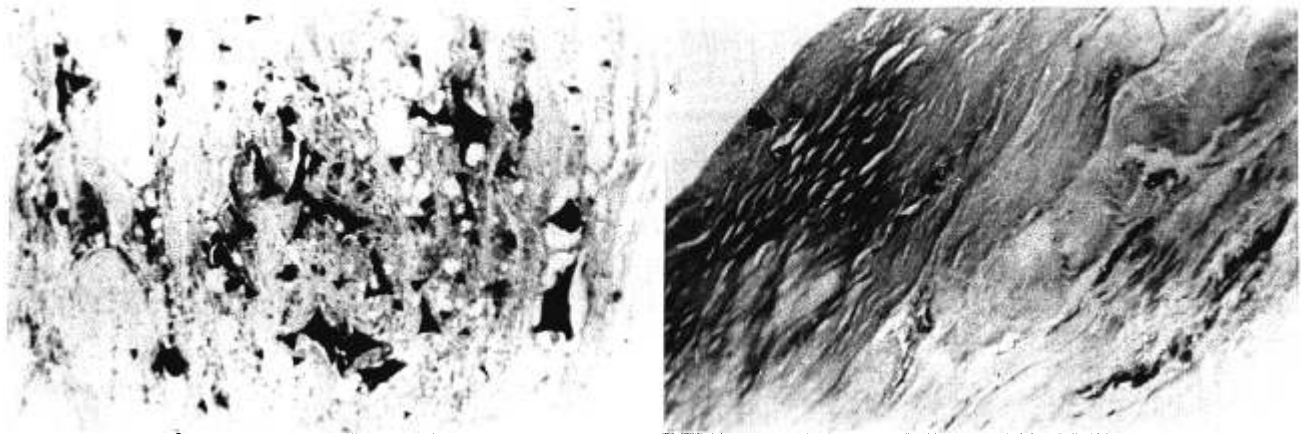


FIG. 12. (Left) Despite the abundance of foam in a capsule from a 5-year implant (Giemsa stain; $\times 100$), visibly intact foam could not be recovered after enzymatic digestion of the capsule. (Right) In contrast, a 19-month capsule (Giemsa stain; $\times 100$) shows absence of a foreign-body inflammatory response and a paucity of foam particles (arrow).

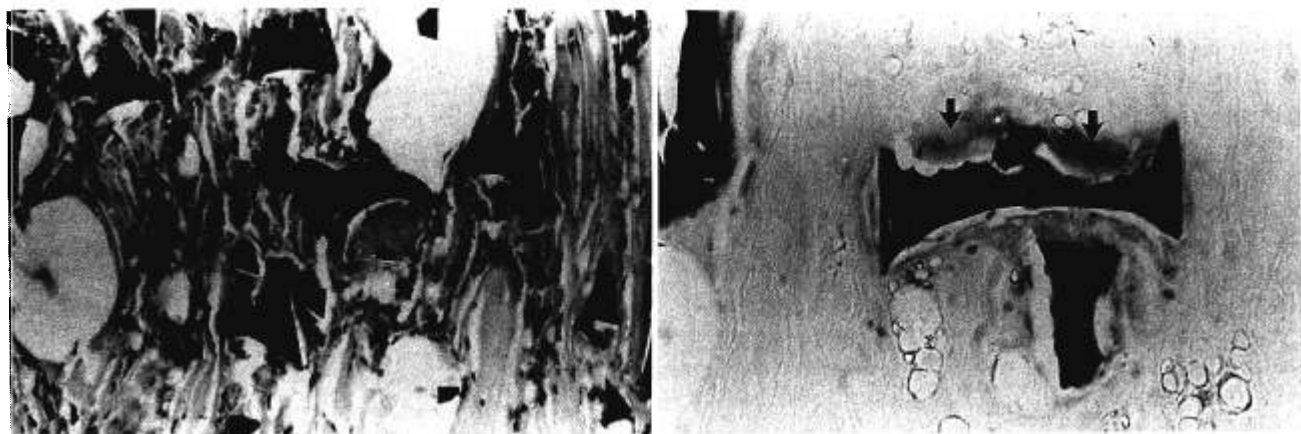


FIG. 13. (Left) Multinucleated giant cell (arrow) seen adjacent to a fractured foam particle (Giemsa stain; $\times 250$). (Right) Scalloping of polyurethane strut from multinucleated giant cells (arrows) is also evident (Giemsa stain; $\times 400$).

based on the probability of retrieving intact polyurethane foam. Polyurethane strut width measurements using scanning electron microscopy

provided us with an alternate, objective, albeit indirect, method of quantifying *relative* biodegradation of polyurethane. Again, this was limited

to the implants with intact foam recovered, but it appears that the group of implants that developed capsular contracture or became infected showed an increased rate of polyurethane biodegradation, as noted by the thinner struts, as compared with implants removed for other reasons. We can surmise that the increased stress forces produced during the process of capsular contraction or the increased number and activity of inflammatory cells during an infection could accelerate the degradative process. Alternatively, some individuals may be "fast" biodegraders and thus at a greater risk for early capsular contracture. We did not test whether the observed differences in polyurethane strut widths were statistically significant, because the implants that did not have intact foam had a "missing value" for the polyurethane strut width. This would cause an overestimation of the width and thus bias the results because only the implants with intact foam would be considered. Hence these relationships are at best a conjecture that requires further investigation.

As far as we are aware, this is the largest series of human polyurethane foam-covered breast "explants" that have been analyzed specifically to look at polyurethane biodegradation. We chose to analyze multiple random capsule biopsies rather than digest the entire capsule of every implant for three reasons. It was not only cost-effective and more time efficient, but most important, the results we obtained from the capsule biopsies appeared just as representative of what was happening to the foam covering as the results from the implants that underwent total capsule dissolution.

This study is not, however, without its pitfalls. As mentioned earlier, we were analyzing implants from a skewed population of patients. We received far more implants that were removed for capsular contracture or infection than what the overall incidence of these problems in the general population would dictate. The reason, naturally, is that there is a large number of patients who are not having problems with their implants, and thus these implants do not get removed as frequently. The consequence is that we do not get to analyze as many of the successful implants as we would like. Nevertheless, there were enough patients with both complicated and uncomplicated postoperative courses to see some relationships emerge between polyurethane degradation and the reason for implant removal. Clearly, a clinically applicable animal model

needs to be established where, under controlled conditions, the degradation of polyurethane as it relates to capsular contracture, infection, time, metabolic by-products, and other factors can be studied. As to the whereabouts and fate of the degradation products of this polyurethane foam and the possibility of any associated health risks, further clinical and laboratory studies are underway.

CONCLUSION

In conclusion, there is convincing evidence by scanning electron microscopy and light microscopy that polyurethane is degrading. Fissuring, fracturing, and thinning of the polyurethane struts are readily apparent. In the specimens analyzed, the foam had undergone sufficient biodegradation by 3 years to lose its structural integrity and had degraded into smaller particles. The techniques utilized in this study do not make it possible to quantitate accurately the rate of degradation, which is not only variable from patient to patient but from site to site within the same capsule specimen. Factors such as capsular contracture, infection, and time appear to play a role in the biodegradation of polyurethane foam in the human body, and these relationships require further investigation.

Carolyn L. Kerrigan, M.D.

Microsurgical Research Laboratories L4.65

Royal Victoria Hospital

687 Pine Avenue, West

Montreal, Quebec, Canada H3A 1A1

ACKNOWLEDGMENTS

We would like to thank Dr. Serge Jothy, from the Department of Pathology, at the Royal Victoria Hospital, for his suggestions concerning the capsule tissue dissolution process and the polyurethane foam staining technique, and Dr. John Sampalis, from the Departments of Surgery and Epidemiology and Biostatistics, at McGill University, for the statistical analyses. We also would like to thank Helen Campbell, from the Department of Mining and Metallurgical Engineering, at McGill University, for instructing the first author in the technique of scanning electron microscopy. A special appreciation goes to all the surgeons, too numerous to mention individually, who contributed prostheses, and we are very grateful but sympathetic to the women who were unfortunate enough to require secondary surgery but without whom this study would not have been possible.

REFERENCES

1. Ashley, F. L. A new type of breast prosthesis: Preliminary report. *Plast. Reconstr. Surg.* 45: 421, 1970.
2. Dick, G. O., and Brown, S. A. Breast reconstruction using modified tissue expansion. *Plast. Reconstr. Surg.* 77: 613, 1986.
3. Hester, T. R., Jr. The Polyurethane-Covered Mam-

- mary Prosthesis: Facts and Fiction. In John Bostwick, 3d (Ed.), *Perspectives in Plastic Surgery*, Vol. 2. St. Louis: Quality Medical Publishers, 1988. Pp. 135-169.
4. Melmed, E. P. Polyurethane implants: A 6-year review of 416 patients. *Plast. Reconstr. Surg.* 82: 285, 1988.
 5. Hester, T. R., Jr., Nahai, F., Bostwick, J., and Cukic, J. A 5-year experience with polyurethane-covered mammary prostheses for treatment of capsular contracture, primary augmentation mammoplasty, and breast reconstruction. *Clin. Plast. Surg.* 15: 569, 1988.
 6. Pennisi, V. R. Polyurethane-covered silicone gel mammary prosthesis for successful breast reconstruction. *Aesthetic Plast. Surg.* 9: 73, 1985.
 7. Brand, K. G. Foam-covered mammary implants. *Clin. Plast. Surg.* 15: 533, 1988.
 8. Picha, G. J., Goldstein, J. A., and Stohr, E. Natural-Y Meme polyurethane versus smooth silicone: Analysis of the soft-tissue interaction from 3 days to 1 year in the rat animal model. *Plast. Reconstr. Surg.* 85: 903, 1990.
 9. Rigdon, R. H. Local reaction to polyurethane: A comparative study in the mouse, rat, and rabbit. *J. Biomed. Mater. Res.* 7: 79, 1973.
 10. Lilla, J. A., and Vistnes, L. M. Long-term study of reactions to various silicone breast implants in rabbits. *Plast. Reconstr. Surg.* 57: 637, 1976.
 11. Smahel, J., Schneider, K., and Donski, P. Bizarre implants for augmentation mammoplasty: Long term human reaction to polyethylene strips. *Br. J. Plast. Surg.* 30: 287, 1977.
 12. Dunaif, C. B., Stubenbord, W. T., and Conway, H. Observations on subcutaneously implanted polyetherurethane sponge in mice. *Surg. Gynecol. Obstet.* 117: 454, 1963.
 13. Smahel, J. Tissue reactions to breast implants coated with polyurethane. *Plast. Reconstr. Surg.* 61: 80, 1978.
 14. Wagner, H., Beller, F. K., and Pfautsch, M. Electron and light microscopy examination of capsules around breast implants. *Plast. Reconstr. Surg.* 60: 49, 1977.
 15. Slade, C. L., and Peterson, H. D. Disappearance of the polyurethane cover of the Ashley Natural-Y prosthesis. *Plast. Reconstr. Surg.* 70: 379, 1982.
 16. Ratner, B. D., Gladhill, K. W., and Horbett, T. A. Analysis of in vitro enzymatic and oxidative degradation of polyurethanes. *J. Biomed. Mater. Res.* 22: 509, 1988.
 17. Marchant, R. E., Anderson, J. M., Phua, de K., and Hiltner, A. In vivo biocompatibility studies: II. Biomer: Preliminary cell adhesion and surface characterization studies. *J. Biomed. Mater. Res.* 18: 309, 1984.
 18. Hennig, E., John, A., Zartnack, F., et al. Biostability of polyurethanes. *Int. J. Artif. Organs* 11: 416, 1988.
 19. Kerrigan, C. L. Report on the Meme Breast Implant. Prepared for the Minister of Health and Welfare, Canada, The Honorable Perrin Beatty, May 1989.
 20. Stokes, K. B. Polyether polyurethanes: Biostable or not. *J. Biomater. Appl.* 3: 228, 1988.
 21. Pinchuk, L., Martin, J. B., Jr., Esquirel, M. C., et al. The use of silicone/polyurethane graft polymers as a means of eliminating surface cracking of polyurethane prostheses. *J. Biomater. Appl.* 3: 260, 1988.
 22. Szycher, M., and Siciliano, A. A. Polyurethane-covered mammary prosthesis: A nine-year follow-up assessment. *J. Biomater. Appl.* 5: 282, 1991.